Industrial Internship Report on

”File Organizer using Python”

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time.

My project was "File Organizer using Python". This project automatically organizes files into different folders based on their file extensions such as Images, Videos, Documents, Audio, and Applications. It was developed using Python's OS and shutil modules to simplify file management and reduce manual effort.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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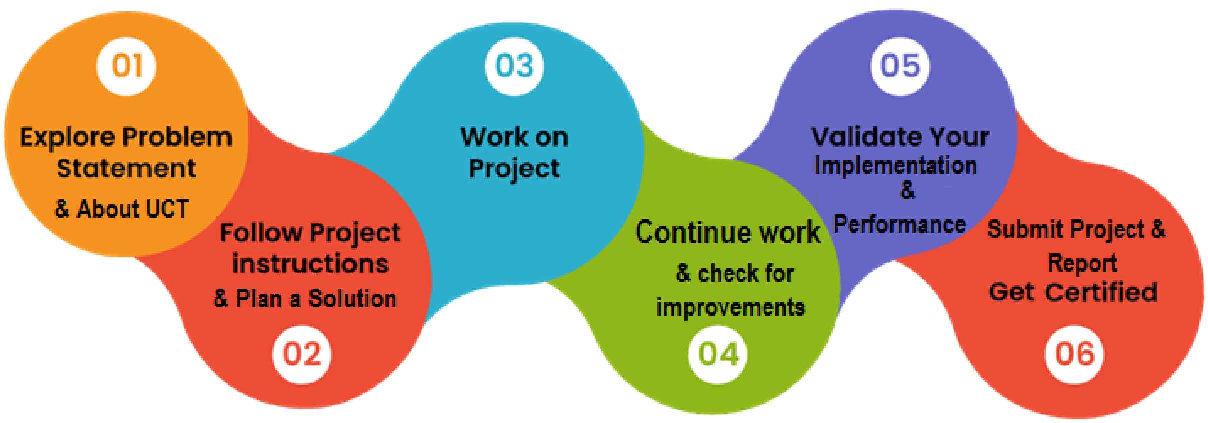
1 Preface

This report presents the work completed during my six-week Python Programming Internship conducted by upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd.

During the internship, I learned the basics of Python programming through weekly learning modules, quizzes, and practical exercises. I also developed a **File Organizer using Python**, which automatically organizes files into different folders based on their file extensions



Overall, this internship helped me improve my understanding of Python programming and gave me hands-on experience in developing a simple real-world project. I would like to thank upskill Campus, UniConverge Technologies Pvt. Ltd., and everyone involved in this internship for providing this valuable learning opportunity.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

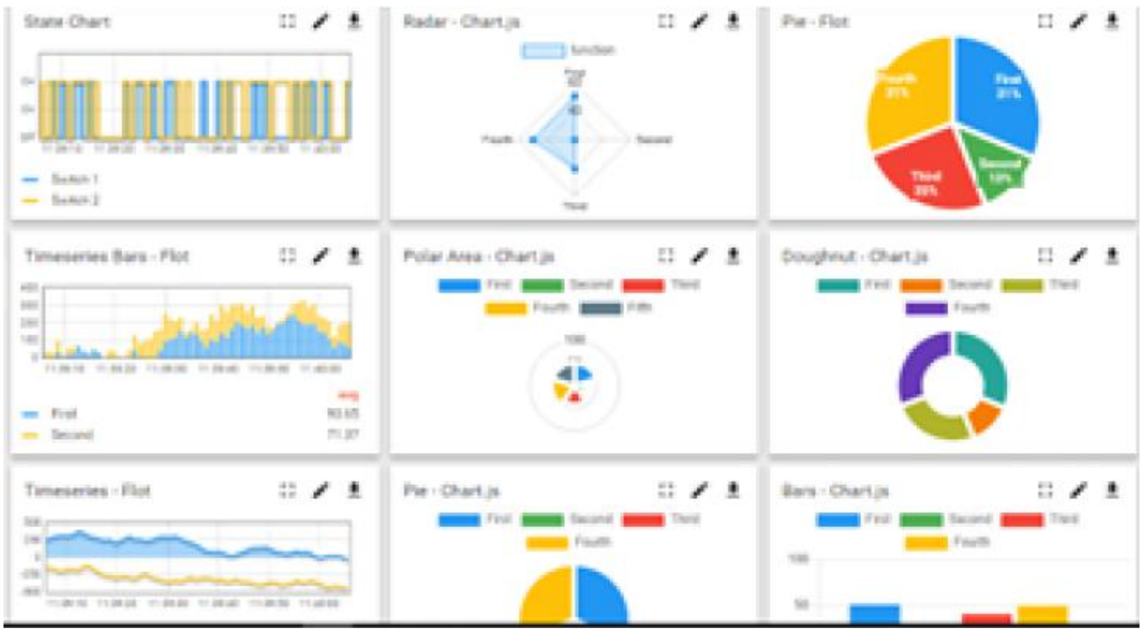
i. UCT IoT Platform (uct Insight)

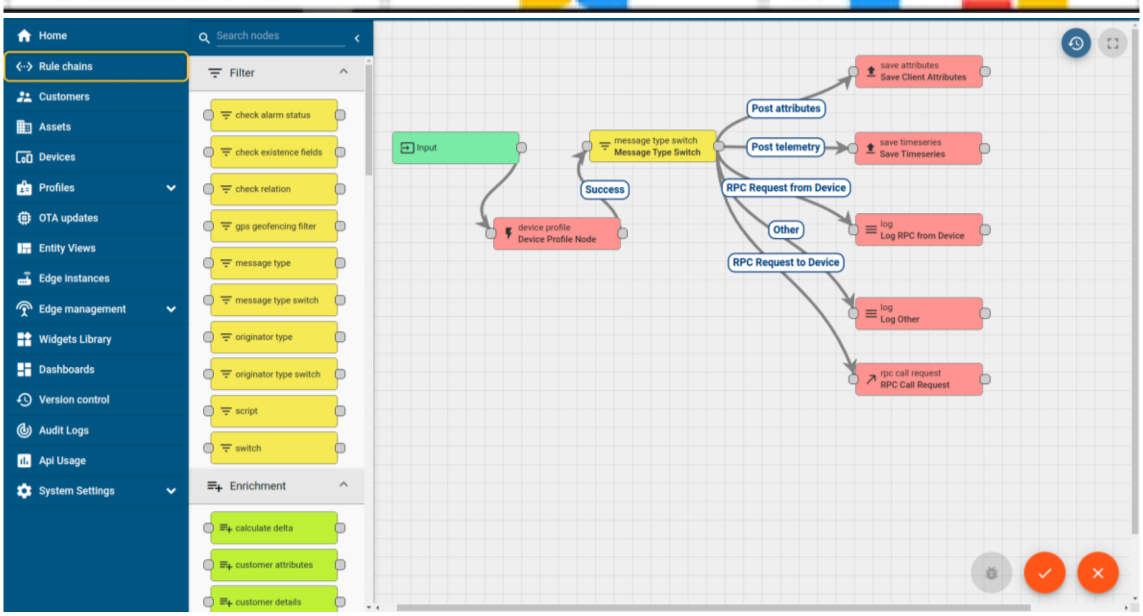
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

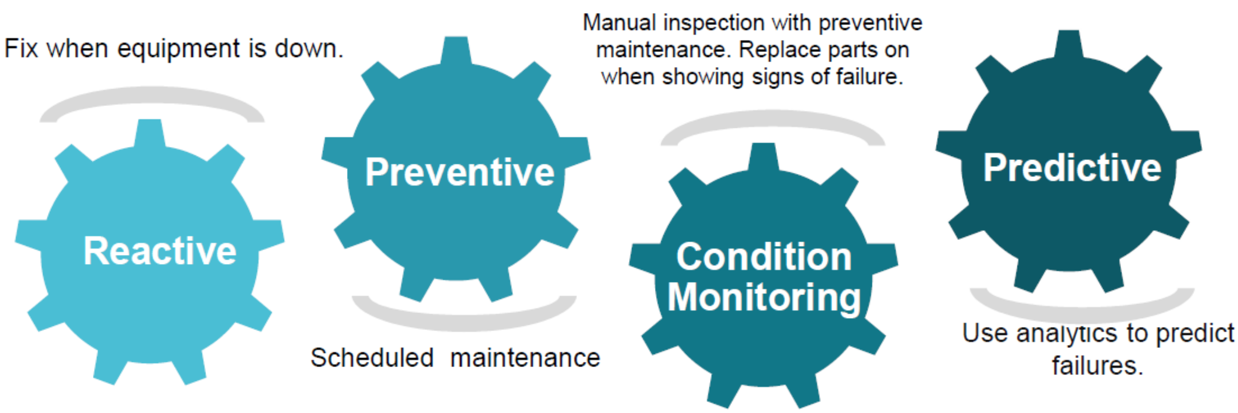
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

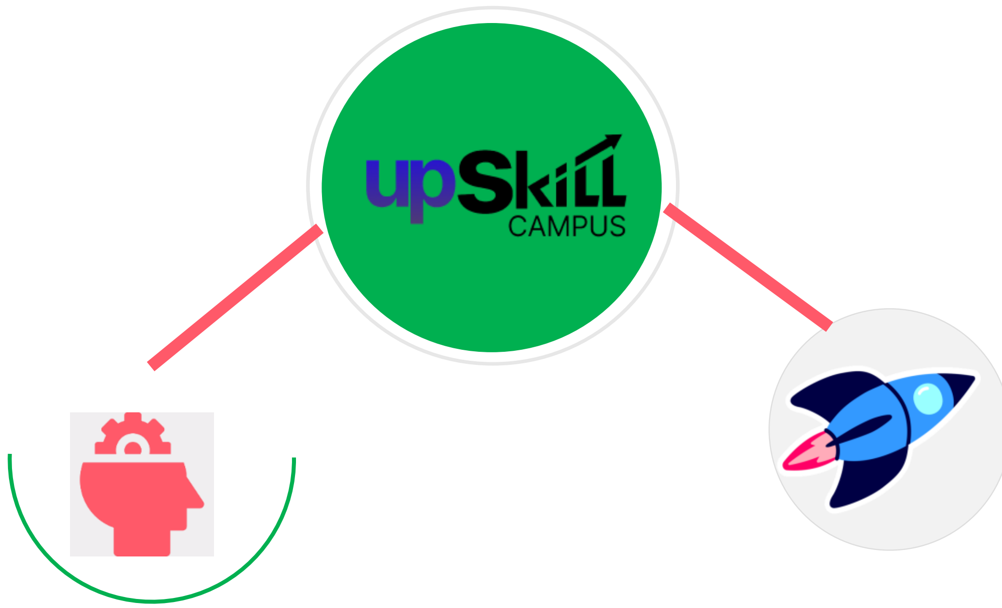
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



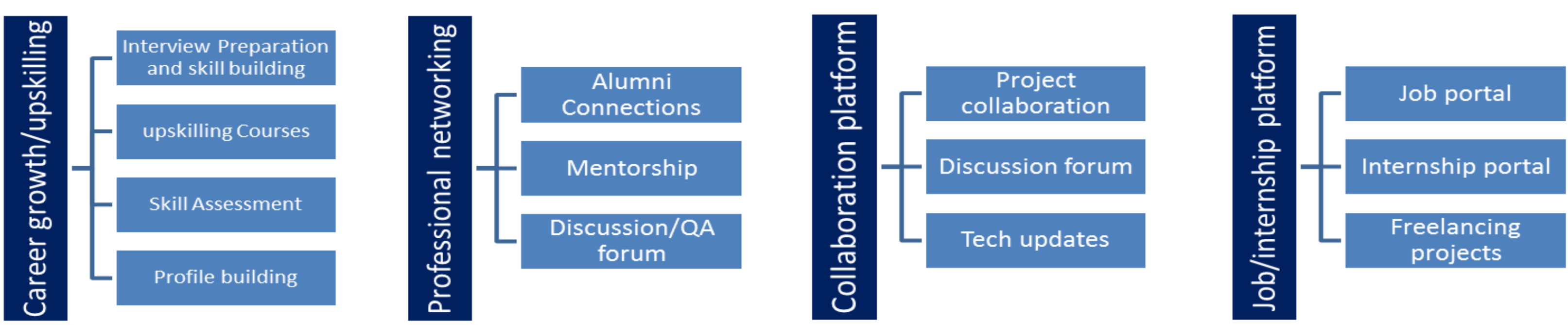
2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Python Official Documentation – https://docs.python.org/

[2] NumPy Official Documentation – https://numpy.org/doc/

[3] Pandas Official Documentation – https://pandas.pydata.org/docs/

[4] Upskill Campus Learning Resources

[5] UniConverge Technologies Internship Materials

2.6 Glossary

Terms	Acronym
Python	-
Operating System	OS
NumPy	Numerical Python
Pandas	PD
Integrated Development Environment	IDE

3 Problem Statement

Managing a large number of files manually can be time-consuming and confusing, especially when different file types are stored in the same folder. Finding specific files becomes difficult if they are not organized properly.

The objective of this project is to develop a **File Organizer using Python** that automatically sorts files into different folders based on their file extensions, such as Images, Videos, Documents, Audio, and Applications. This helps in keeping files organized and reduces manual effort.

Problems Faced

- Different types of files are stored in a single folder.
- Finding a required file takes more time.
- Manual file organization is repetitive and time-consuming.
- Files are not categorized automatically.
- A messy folder reduces productivity.

Objectives of the Project

- Organize files automatically based on their file extensions.
- Create folders for different file categories if they do not exist.
- Reduce manual effort in managing files.
- Keep the working directory clean and organized.
- Provide a simple automation solution using Python.

4 Existing and Proposed solution

Existing Solution

Currently, files are organized manually by the user. When different file types are stored in the same folder, the user has to create folders manually and move each file one by one.

Limitations of Existing Solution:

- Time-consuming for large number of files.
- Repetitive manual work.
- Chances of placing files in the wrong folder.
- Difficult to keep folders organized regularly.

Proposed Solution

The proposed solution is a Python-based File Organizer that automatically scans a selected folder, identifies file extensions, creates category folders if required, and moves files into their respective folders.

Features of the Proposed Solution:

- Automatically scans files in a folder.
- Detects file type using file extensions.
- Creates folders automatically if they do not exist.
- Moves files to their respective folders.
- Supports common file types like Images, Videos, Documents, Audio, Applications, and Others.

Advantages of the Project

- Saves time by reducing manual work.
- Keeps files organized automatically.
- Easy to use and understand.

4.1 Code submission (Github link) :
https://github.com/devwrat05/upskillcampus/blob/main/File_Organizer_in_Python.py

4.2 Report submission (Github link) : first make placeholder, copy the link.

5 Proposed Design/ Model

The File Organizer follows a simple workflow. The program reads all files from the selected folder, checks their file extensions, creates folders if required, and moves each file to its respective category. The design is simple, easy to understand, and suitable for beginners learning Python automation.

5.1 High Level Diagram:

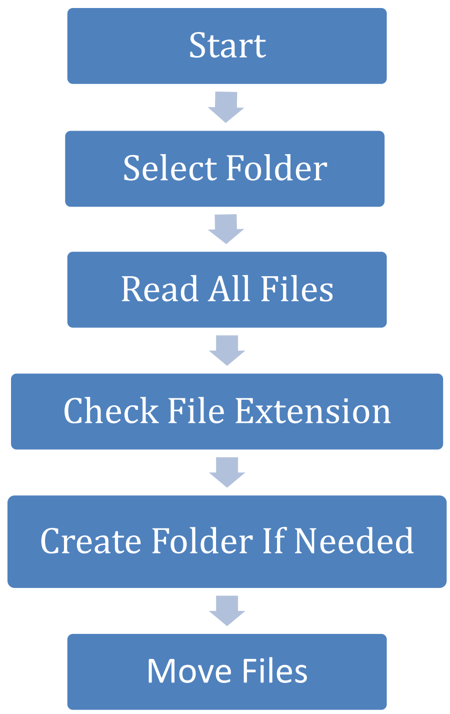
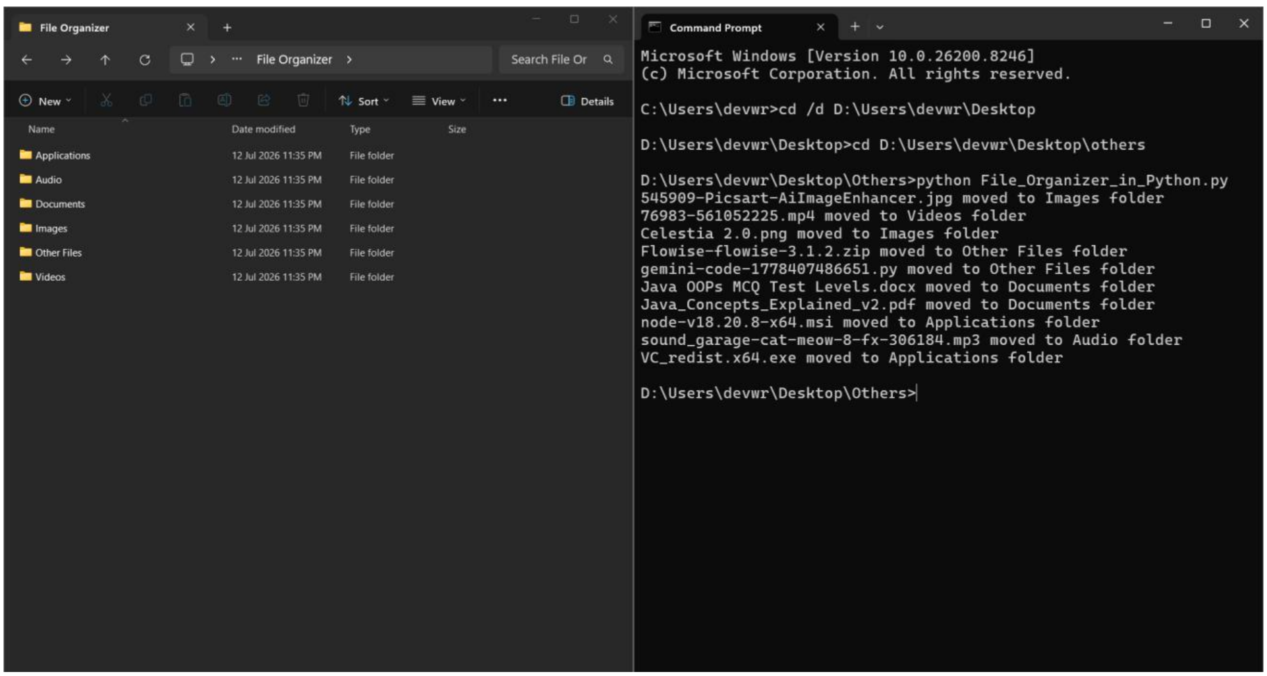


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces :

This project is a command-line application. The user provides the folder path in the Python program, and the output is displayed in the terminal. After execution, categorized folders are created automatically in the selected directory.



6. Performance Test

The File Organizer project was tested using different types of files stored in a single folder. The objective of testing was to verify whether the program correctly identifies file extensions, creates folders, and moves files to their respective categories. The program worked successfully for all supported file types.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Images (.jpg, .png, .jpeg)	Moved to Images folder	Pass
Videos (.mp4)	Moved to Videos folder	Pass
Documents (.pdf, .docx)	Moved to Documents folder	Pass
Audio (.mp3)	Moved to Audio folder	Pass
Applications (.exe, .msi)	Moved to Applications folder	Pass
Other (.zip, .py)	Moved to Other Files folder	Pass

6.2 Test Procedure

1. Created a folder containing different types of files.
2. Updated the folder path in the Python program.
3. Executed the program.
4. Verified that category folders were created automatically.
5. Checked whether all supported files were moved to the correct folders.

6.3 Performance Outcome

- All supported files were successfully organized into their respective folders.
- The required folders were created automatically.
- The program completed the task without any runtime errors during testing.
- The output matched the expected results for all tested file types.

7 My learnings

During this internship, I gained a good understanding of the basics of Python programming. I learned concepts such as variables, conditional statements, loops, functions, file handling, NumPy, and Pandas. Weekly quizzes helped me revise these concepts and improve my understanding.

Developing the **File Organizer** project gave me practical experience in using Python modules like **os** and **shutil**. I also learned how to identify problems, write simple logic, test programs, and debug errors.

This internship improved my confidence in writing Python programs and motivated me to continue learning more advanced topics in the future. The knowledge gained during this internship will help me in future academic projects, internships, and career opportunities related to programming.

8 Future work scope

The current project is a beginner-level file organization system. In the future, several improvements can be added to make it more useful and user-friendly.

Some possible future enhancements are:

- Develop a graphical user interface (GUI) for easier use.
- Allow users to choose the folder through a file browser.
- Add support for more file types and custom categories.
- Automatically organize files at regular intervals.
- Display a summary of organized files after execution.
- Improve error handling and logging.

Overall, this project can be extended into a more advanced file management application by adding additional automation features.